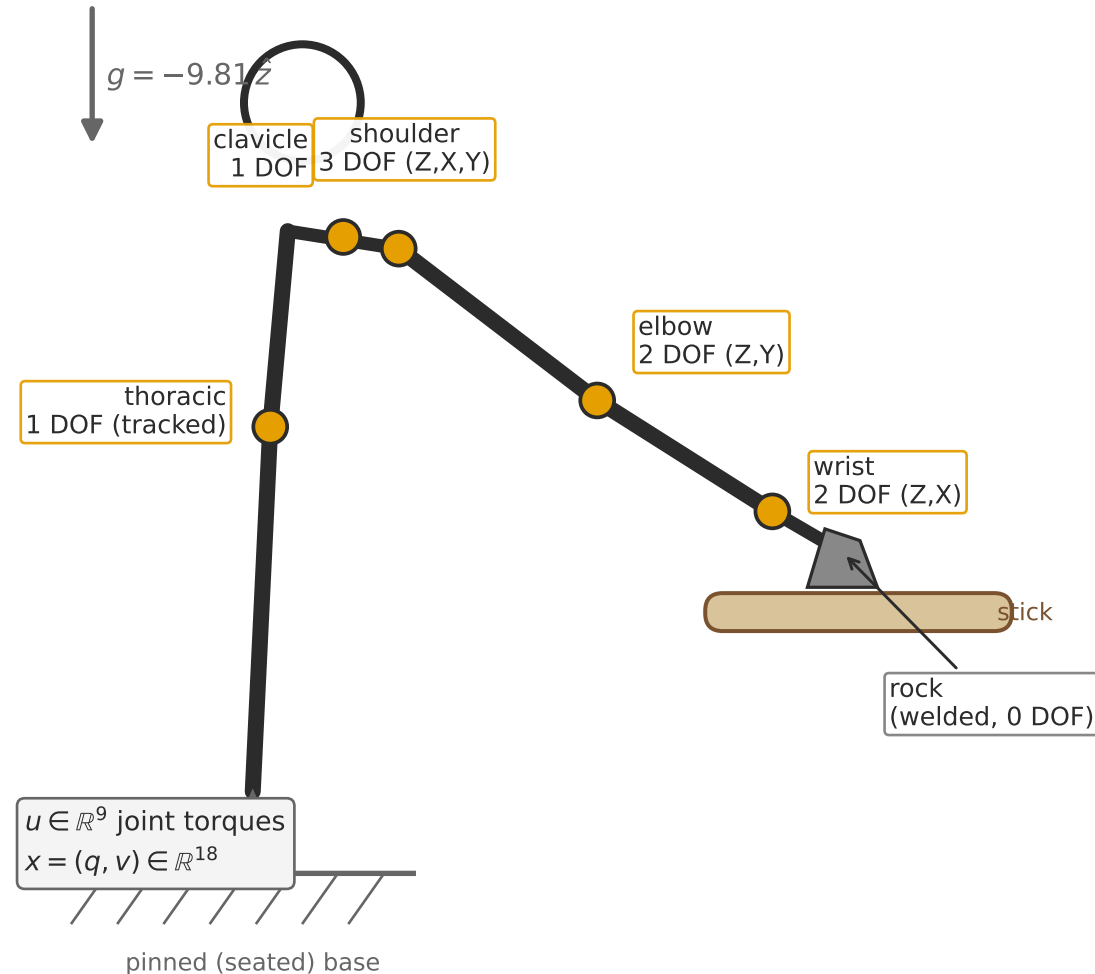


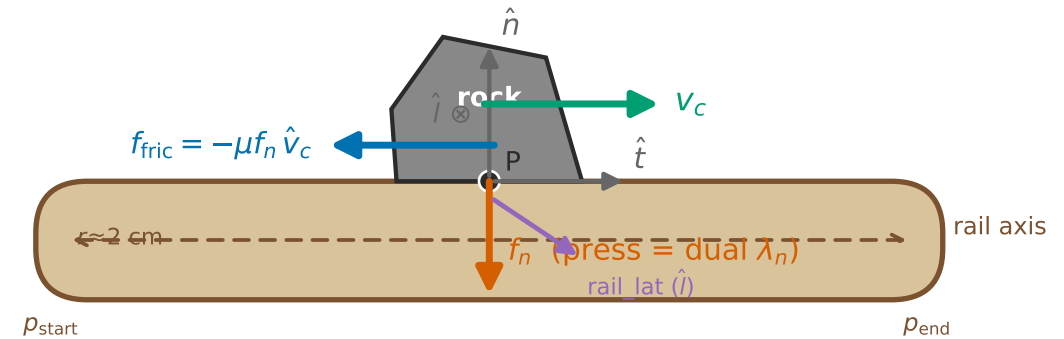
The physical model — seated human scraping a stone tool along a wooden stick

(a) 9-DOF seated upper limb



(b) rock-stick contact (1-D moving rail)

1-D holonomic: normal pinned, tool slides along $\hat{t}$



(c) the model, in equations

Constrained dynamics:

$$M(q)\ddot{q} + h(q, \dot{q}) = \tau_{\text{act}} + J_c(q)^T \lambda$$

Actuation split (control $\neq$ muscle torque):

$$\tau_{\text{act}} = u + J_c(q)^T f_{\text{fric}}$$

Coulomb friction ($\mu = 0.3$):

$$f_{\text{fric}} = -\mu f_n v_c / \|v_c\|$$

Press is a free contact dual (recovered):

$$f_n = \lambda_n$$

Contact features:

$$\frac{1}{2}(f_n - f_{\text{tgt}})^2, \quad \frac{1}{2}(f_n - F_{\text{max}})^2 \Rightarrow f_n^* = \frac{w_{\text{cap}}}{w_{\text{pf}} + w_{\text{cap}}} F_{\text{max}}$$

Time-varying IRL cost:

$$C(t) = \sum_i w_i(t) \phi_i(x, u), \quad w(t) = \sum_k B_k(t) \theta_k$$